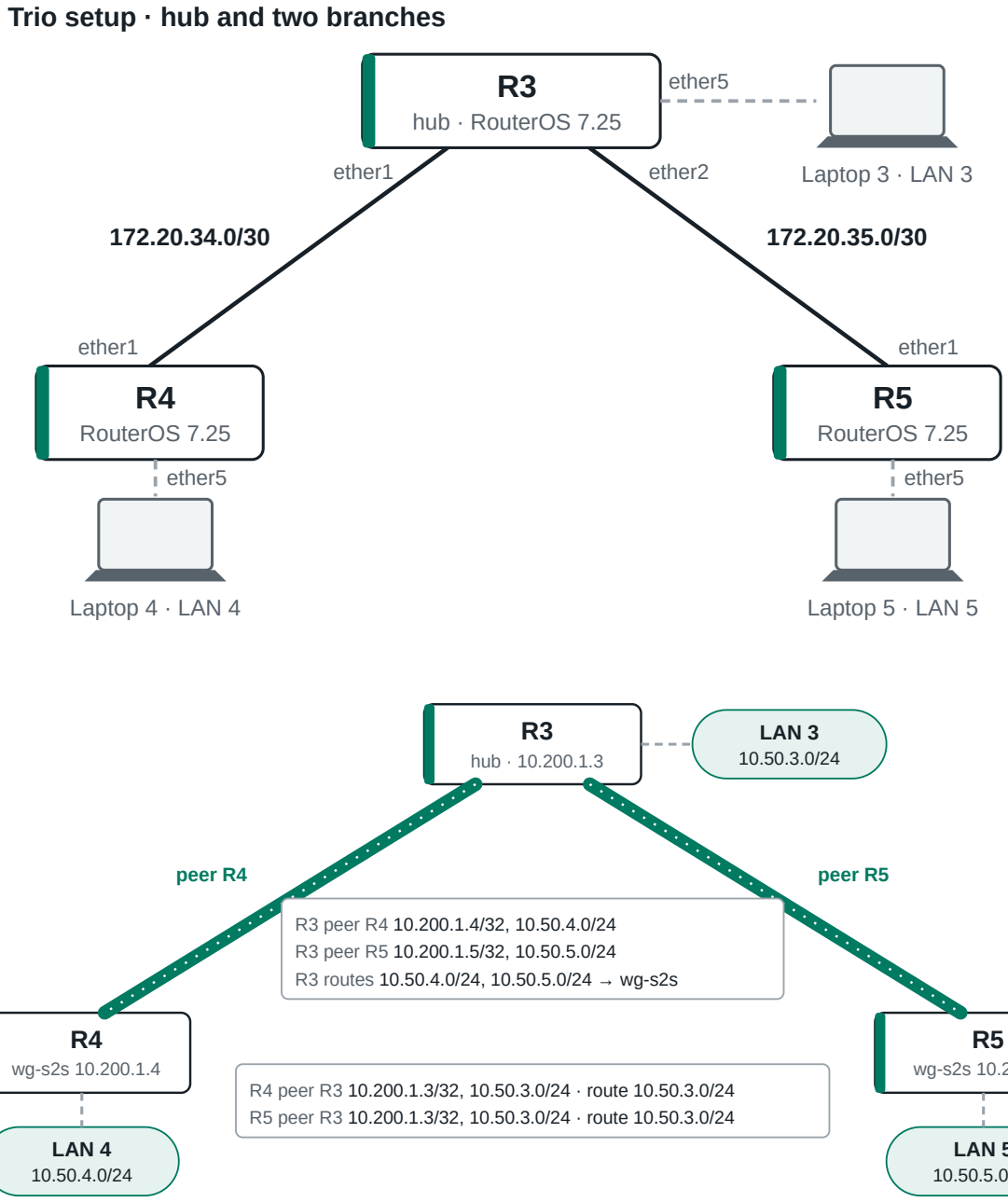
WireGuard Site-to-Site VPN — Trio setup

Train the Trainer · RouterOS v7 · Student handout · routers R3 (hub), R4 and R5

1. Topology



2. Addressing plan

Router	Transport	Tunnel wg-s2s	Site LAN (ether5)
R3 — hub	ether1 172.20.34.1/30, ether2 172.20.35.1/30	10.200.1.3/24	10.50.3.1/24
R4 — branch	ether1 172.20.34.2/30	10.200.1.4/24	10.50.4.1/24
R5 — branch	ether1 172.20.35.2/30	10.200.1.5/24	10.50.5.1/24

WireGuard port **13231** · interface **wg-s2s** · laptop on ether5 (DHCP)

3. Tasks

3.0 Prepare your router

Board config you need later? Back it up first and copy both files to your laptop:

```
/system backup save name=before-ttt
/export file=before-ttt
```

Download **start-*Rn*.rsc** for your router from github.com/111REB579/TTT_2026 and drag it into WinBox *Files*. Then:

```
/system reset-configuration skip-backup=yes keep-users=yes no-defaults=yes run-after-reset=start-<my router>.rsc
```

Renew the DHCP lease on your laptop (10.50.*n*.x) and reconnect WinBox to 10.50.*n*.1.

```
/ip address print
/ping <partner transport address> count=3
```

3.1 Create the WireGuard interface

```
/interface wireguard add name=wg-s2s listen-port=13231
/interface wireguard print
```

3.2 Tunnel address and key swap

```
/ip address add address=<my tunnel address>/24 interface=wg-s2s
```

Swap public keys: R3 needs the keys of R4 and R5; R4 and R5 need the key of R3. Compare the last four characters.

3.3 Allow WireGuard on the transport

```
/ip firewall filter add chain=input action=accept protocol=udp dst-port=13231 in-interface-list=WAN comment="allow WireGuard" place-before=[find comment="defconf: drop all not coming from LAN"]
/ip firewall filter print where chain=input
```

Always check that the port is open: "allow WireGuard" must be **above** "drop all not coming from LAN".

3.4 Add the peer

R3: twice, for R4 and for R5.

```
/interface wireguard peers add interface=wg-s2s name=<partner> public-key="<partner public key>" endpoint-address=<partner transport address> endpoint-port=13231 allowed-address=<partner tunnel address>/32,<partner site LAN>
/interface wireguard peers print detail
```

Checkpoint 1 — handshake and tunnel

```
/ping <partner tunnel address> count=3
/interface wireguard peers print detail
```

Pass: replies; `last-handshake` has a value; `rx` and `tx` grow. R3: for both branches.

3.5 Route the remote LAN

R3: two routes, one per branch LAN.

```
/ip route add dst-address=<partner site LAN> gateway=wg-s2s
/ip route print where gateway=wg-s2s
```

Checkpoint 2 — LAN to LAN, both directions

From the laptop:

```
ping <partner LAN gateway>
```

From the router:

```
/ping <partner LAN gateway> src-address=<my LAN gateway> count=3
```

Pass: replies in both directions.

4. Self-check

	I have seen that …	Where	✓
1	my laptop has a 10.50.*n*.x address and my partner's transport address answers	3.0	☐
2	wg-s2s exists and carries my tunnel address	3.1–3.2	☐
3	"allow WireGuard" is above "drop all not coming from LAN"	3.3	☐
4	the public key in my peer entry ends with the same 4 characters as my partner's own key	3.4	☐
5	`last-handshake` has a value; `rx` and `tx` grow	Checkpoint 1	☐
6	my partner's tunnel address answers	Checkpoint 1	☐
7	the route to my partner's LAN is active (`A`)	3.5	☐
8	my laptop reaches my partner's LAN, and my partner's laptop reaches mine	Checkpoint 2	☐

R3 ticks lines 4–8 only when they hold for both R4 and R5.

5. Done early? (optional)

Branch to branch through the hub — on R4 and R5 only (*o* = the other branch: 5 on R4, 4 on R5):

```
/interface wireguard peers set [find name=R3] allowed-
address=10.200.1.3/32,10.50.3.0/24,10.200.1.<o>/32,10.50.<o>.0/24
/ip route add dst-address=10.50.<o>.0/24 gateway=wg-s2s
```

6. If something does not work

1. **Transport:** does the partner's transport address answer?
2. **Handshake:** `tx` grows, `rx` stays 0? Check the partner's input rule, the endpoint address and the public key.
3. **Tunnel:** does the partner's tunnel address answer?
4. **Route:** "Destination network unreachable" from your own router? The route is missing.
5. **allowed-address:** tunnel fine, LAN ping times out? The partner's allowed-address does not list your LAN.
6. **Hub:** "allowed-address conflicts other peer"? Each prefix belongs to exactly one peer.

Did not finish? In the GNS3 student project (login `admin` / `wg-lab`) the routers are in the state after 3.0: continue with 3.1. To start a router over: `/system reset-configuration skip-backup=yes keep-users=yes no-defaults=yes run-after-reset=start-<router>.rsc`